PH784 Advanced Social Epidemiology: Methods

Week 2: Causal Inference & Social Epi

Koichiro Shiba

kshiba@bu.edu

Boston University School of Public Health

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Structure

1. Causal Inference Intro
    - General Overview
2. Challenges in Social Epi
    - Consistency

Social Epidemiologic Questions

- **Assessing/Monitoring Health Inequalities**
  - → (Mostly) descriptive
- **Social Determinants of Health**
  - → Individual, inter-personal, neighborhood, macro-level social factors
  - → Causal
    - ▫ Today's focus

> **We will discuss the intersection of these two in Week 5**

Goal of Causal Inference

Potential outcome framework

- Quantitative
- Estimating the causal effect of *a hypothetical intervention*
  - → Even in observational studies
  - → e.g., $E[Y^{a=1} - Y^{a=0}]$
    - ▫ Y^a: Potential outcome under a binary treatment $A = a$
      - ▫ More generally, $E[Y^a - Y^{a^*}]$
    - ▫ "Difference in mean outcome had everyone been exposed vs had everyone been unexposed"
- Not establishing causality

Conceptualizing a Hypothetical Intervention (in Social Epi)?

- ALL causal inference is doing this
    - → implicitly
    - → or explicitly
        - e.g., "Target trial emulation" (Hernán and Robins 2016)

Benefits of explicitly conceptualizing a hypothetical intervention

1. Rigorous study design
2. Clearer causal questions & interpretation
    - Greater relevance to the real world

Conceptualizing a Hypothetical Intervention (in Social Epi)?

Common critiques

- "It narrows the scope of science!"
  - → Not negating other approaches for causality/scientific discoveries
- "Impossible with observational data!" & "Only correlational!"
  - → Sure, it is more challenging. But isn't it what you are doing anyway?
- "Not all social factors can be intervened!"
  - → Why would you be interested in effects of something you cannot change?
    - *More on this later today*

Interpret Estimated Effects of SDoH

- Social factors sometimes have small “effect sizes” compared to genetic/clinical factors

Why should we care SDoH, even those with small “effect sizes”?

- Upstream/root cause
- Takes time for effects to manifest
- Greater population impacts

“Effect Size” and Population Impacts

Hypothetical scenario

- Effect of a binary (protective) social exposure A on a disease Y: RR=0.98
  - → How many cases can be prevented by assigning this exposure to unexposed individuals?
  - → Baseline outcome risk among the unexposed: 10%

# Intervened	# Cases	# Cases if Intervened	# Cases Prevented
1,000	100		
10,000	1,000		
100,000	10,000		

“Effect Size” and Population Impacts

Hypothetical scenario

- Effect of a binary (protective) social exposure A on a disease Y: RR=0.98
  - → How many cases can be prevented by assigning this exposure to unexposed individuals?
  - → Baseline outcome risk among the unexposed: 10%

# Intervened	# Cases	# Cases if Intervened	# Cases Prevented
1,000	100	98	2
10,000	1,000	980	20
100,000	10,000	9,800	200

- Greater population impacts with increasing # of those intervened [1]
  - → What determines this number?

“Effect Size” and Population Impacts

What determines “# Intervened”?

1. Exposure distributions
   - Rare exposures with strong “effects” vs common exposures with modest effects
     - → e.g., genetic mutation vs neighborhood environment
2. Scope of interventions
   - Targeted interventions vs population approaches
     - → e.g., clinical practice vs policies

- Causal $RR = Pr[Y = 1^{a=1}]/Pr[Y = 1^{a=0}]$
  - → “Had everyone been exposed vs unexposed”
    - Ignoring “# Intervened”
    - Change the definition of effects (Week 5) or use population impact metrics (e.g., PAF)

Summary 1

- Causal inference is a task of causal effect estimation
  - → Mathematical definition of causal effects based on the potential outcome framework
  - → Effect of a “hypothetical intervention”
- Consider population-level impacts when interpreting estimated effects of SDoH

Common Approach in SDoH Research

- Regress an outcome of interest on
    - → Social exposure
    - → Adjusting for many covariates
        - Interpret the coefficient as the “effect” of the social variable

> - What effects are we estimating?
> - What are the underlying assumptions?

Formal Causal Inference: Big Picture

Three steps of causal inference

1. Causal Estimand

$E[Y^{a=1}] - E[Y^{a=0}]$

"Causal effect of A on Y"

↓

2. Statistical Estimand

$E[Y|A = 1] - E[Y|A = 0]$

"Association between A and Y"

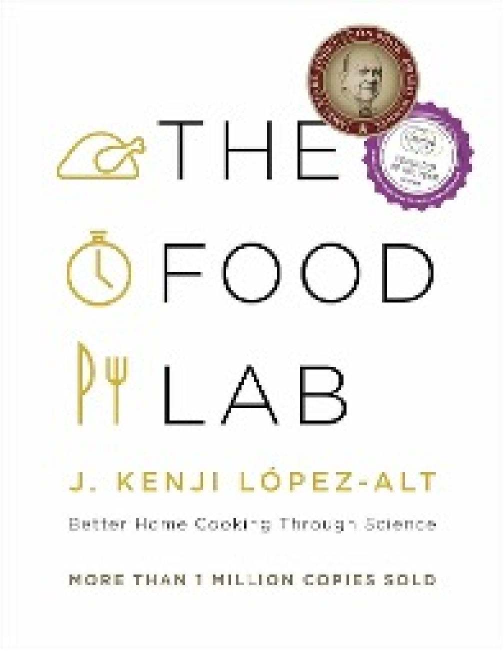
↓

3. Effect Estimate

$E[\widehat{Y|A =} 1] - E[\widehat{Y|A =} 0]$

- General approach for causal inference
  - → Simplified causal roadmap
    - Original paper (Petersen and Laan 2014)
    - Tutorial in E4A Blog post[1]
- You will always get some "numbers" from statistical analyses
  - → Do they have the interpretation that you hope them to have?
  - → Which of these is your "fancy method" improving?
- Challenges in social epi
  - → Mostly estimand and identification

Estimand, Estimator, Estimate



Estimand
(Goal)

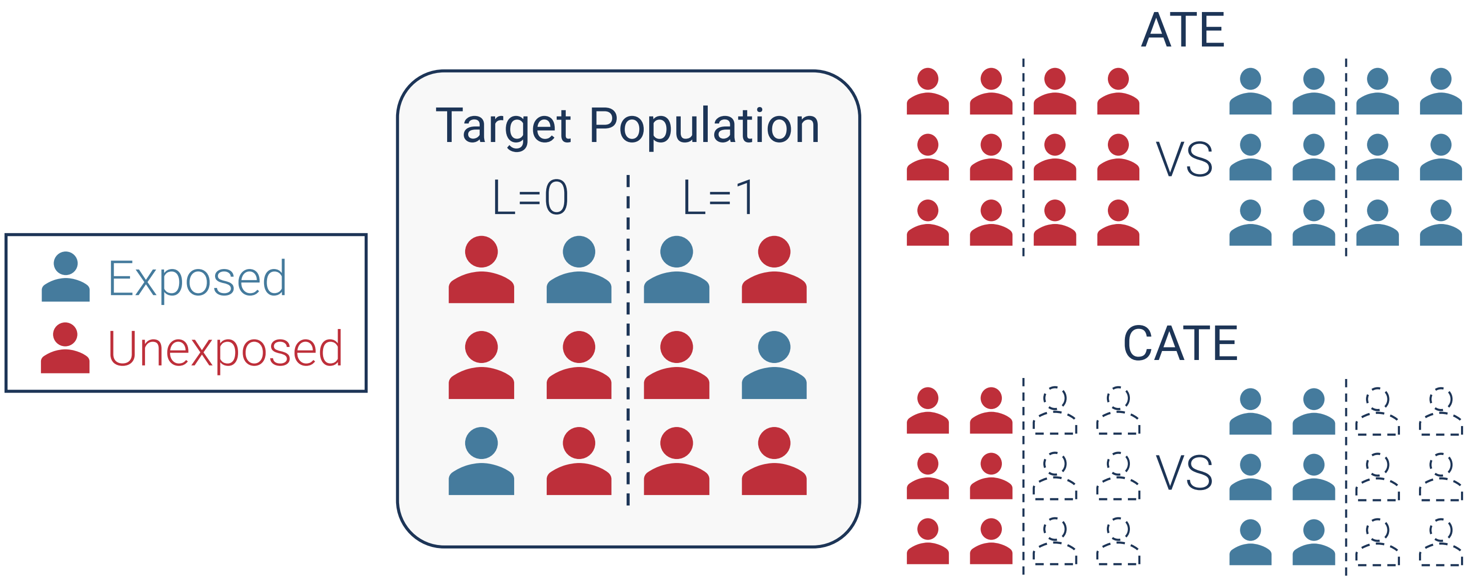
Estimator
(Tool)

Estimate
(Result)

Step 1: Specifying Causal Estimand

Defining a question that you are trying to answer

- **Common causal estimand**
  - → Average treatment effect (ATE): $E[Y^{a=1} - Y^{a=0}]$
  - → Conditional average treatment effect (CATE): $E[Y^{a=1} - Y^{a=0}|L]$



Specifying Causal Estimand

Defining a question that you are trying to answer

- Causal effect: $E[Y^{a=1} - Y^{a=0}]$
  - → Specific to the study population
    - Transportability
  - → Summarized as an expected change in *mean* outcome
    - Can we summarize the outcome differently?
- What is the "hypothetical intervention"?

Causal estimand

Example: the effect of smoking

- Prevalence difference: $Pr[Y^{a=1} = 1] - Pr[Y^{a=0} = 1]$
  - → "Had everyone smoked vs had nobody smoked"
  - → Is this a well-defined hypothetical intervention?

Causal estimand

Example: the effect of smoking

- To specify a well-defined hypothetical intervention, consider:
    - → **What treatment to implement**
        - What type of smoking? (e.g., vaping, tobacco)
        - Dose? (once a week? how many?)
            - Consistency (will be discussed today)
    - → **How to implement**
        - Initiation vs cessation
        - Treatment regimen (e.g., stop smoking now vs forever)

Causal Identification

Where the magic happens

Identification

$$\underbrace{\overset{\textbf{Causal Effect}}{E[Y^{a=1}|L] - E[Y^{a=0}|L]}}_{\text{Counterfactual (unobservable)}} \ggg \underbrace{\overset{\textbf{Statistical Association}}{E[Y|A=1,L] - E[Y|A=0,L]}}_{\text{Factual (observable)}}$$

Identification assumptions

1. Exchangeability
2. Positivity
3. Consistency

Exchangeability

Attributing outcome differences to an exposure

- $E[Y^a|A = 1] = E[Y^a|A = 0]$
  - → Intuition: "Fair comparisons"
  - → "The exposed and the unexposed would have the same outcome had they received the same exposure level"
  - → ~~"The two groups are similar except for their exposure levels"~~

- a.k.a., Ignorability, ~~"no confounding" assumption~~
  - → Can be violated due to confounding, selection bias, measurement errors
    - Is this valid in RCTs?
    - How about in observational studies?

Conditional exchangeability

- $E[Y^a|A=1,L]=E[Y^a|A=0,L]$
  - → Exchangeability within strata of (or conditional on) L

- Interpretation?
  - → "Among those with the same L, the exposed and the unexposed would have the same outcome with the same exposure level"
    - "(Thus, the difference in Y is attributable to A)"
  - → "After adjusting for L, there is no confounding (and other sources of non-exchangeability)"
    - A standard assumption underlying regression adjustment

Positivity

A conceptual & practical issue

- $Pr[A = a|L = l] > 0$
  - → Intuition: "We can compare different exposure levels"
  - → "Within L, there are both exposed and unexposed people"

- Potential violation?
  - → **Structural violation**
    - ▫ "the effect of retirement among newborn babies"
    - ▫ Zero treatment propensity, even with infinite data
      - ▫ Carefully choose target population and define causal estimand
  - → **Random violation**
    - ▫ "In my data, those who are 57 years old and earn $80K per year are all exposed"
    - ▫ No variation in exposure within finite data but can have non-zero propensity in principle
      - ▫ Can be addressed via model interpolation or ensuring common support (e.g., PS matching)

Consistency

- $Y^a = Y$ when $A = a$
  - → Intuition: "The observed outcome Y of the exposed is equal to what the outcome would have been had they been exposed"
  - → Seems obvious? Any potential violation?

- Can be violated when an exposure corresponds to multiple versions of potential interventions
  - → Compound treatment
  - → Versions with different effects
    - Obesity, income, "risk scores", "index", "exercise > 30 minutes/day"

Consistency

- The effect of “aspirin”
  - → Dose? Frequency? With or without food?
- The effect of “a daily tablet of 150 mg of aspirin before breakfast”
  - → With left vs right hand?
    - Same effect
    - Sufficiently well-defined
    - The treatment-variation irrelevance assumption

Consequences of ill-defined exposures

- Unclear actions
- Transportability
- Potentially ill-justified confounding adjustment

Identification Using Assumptions

Identification

$$\underbrace{\overset{\textbf{Causal Effect}}{E[Y^{a=1}|L] - E[Y^{a=0}|L]}}_{\text{Counterfactual (unobservable)}} \ggg \underbrace{\overset{\textbf{Statistical Association}}{E[Y|A=1,L] - E[Y|A=0,L]}}_{\text{Factual (observable)}}$$

Under the identification assumptions, let's connect the counterfactual outcome to the factual outcome

1. $E[Y^{a=1}|L] = E[Y^{a=1}|A=1,L]$
    - conditional exchangeability
2. $E[Y^{a=1}|A=1,L] = E[Y|A=1,L]$
    - consistency

> ! **Unobservable causal estimand is reduced to observable statistical estimand**

Estimation

- How do we estimate $E[Y|A=1,L]-E[Y|A=0,L]$?
  - → "Difference in conditional expected values of Y"
    - This is what regression estimates

Example: BMI by race and gender in NHANES data

▶ Code

Using regression

term	estimate
(Intercept)	26.64
Race1Black	3.14
Gendermale	0.16
Race1Black:Gendermale	-3.6

- The estimate for `Race1Black`: 3.14
  - → Interpretation?
  - → "Estimated association between black race (vs white) and BMI *among female*"

Manual pair-wise comparison

Race1	Gender	mean
Black	female	29.78
Black	male	26.34
White	female	26.64
White	male	26.8

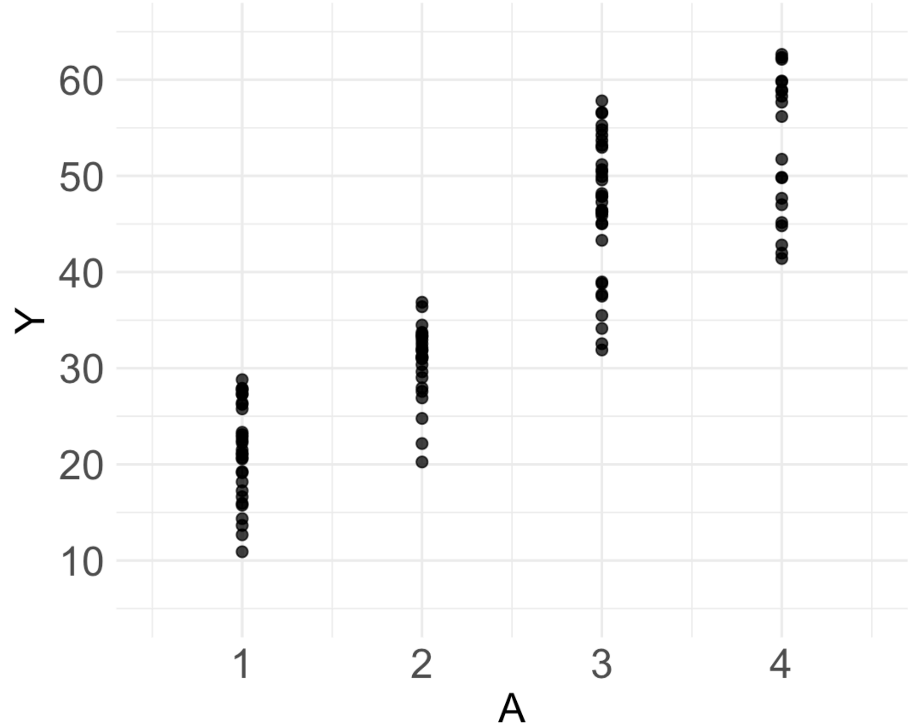
- Difference in BMI between black and white *among female*: $29.78-26.64=3.14$
  - → $E[Y|A=1,L=0]-E[Y|A=0,L=0]$
    - $A=1$ for black and $L=0$ for female.

Estimation

Regression: intuition

$E[Y|A=2]$

▶ Code



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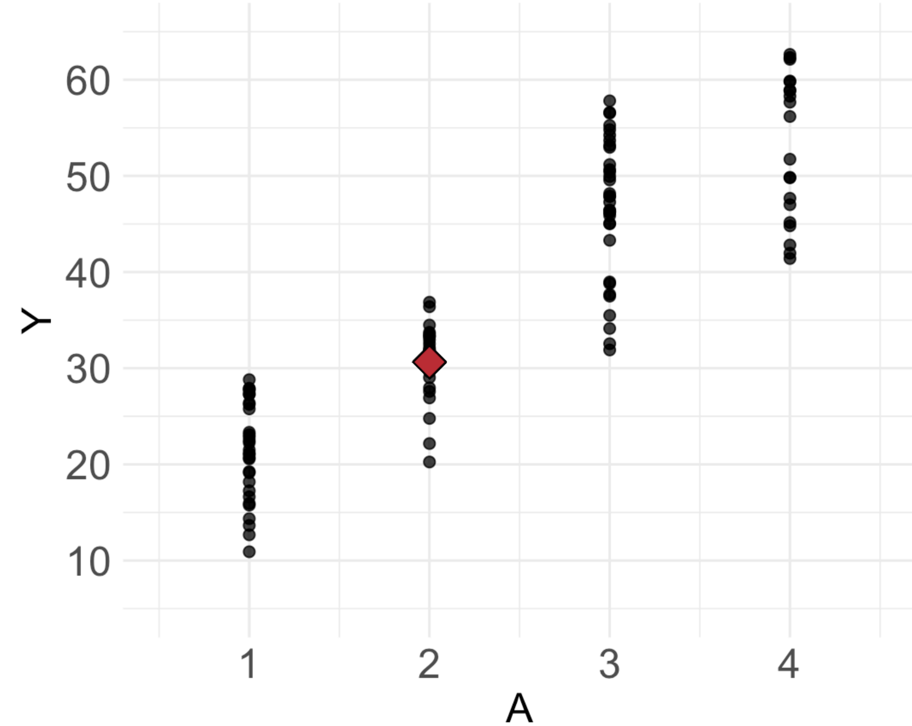

Estimation

Regression: intuition

$E[Y|A=5]$

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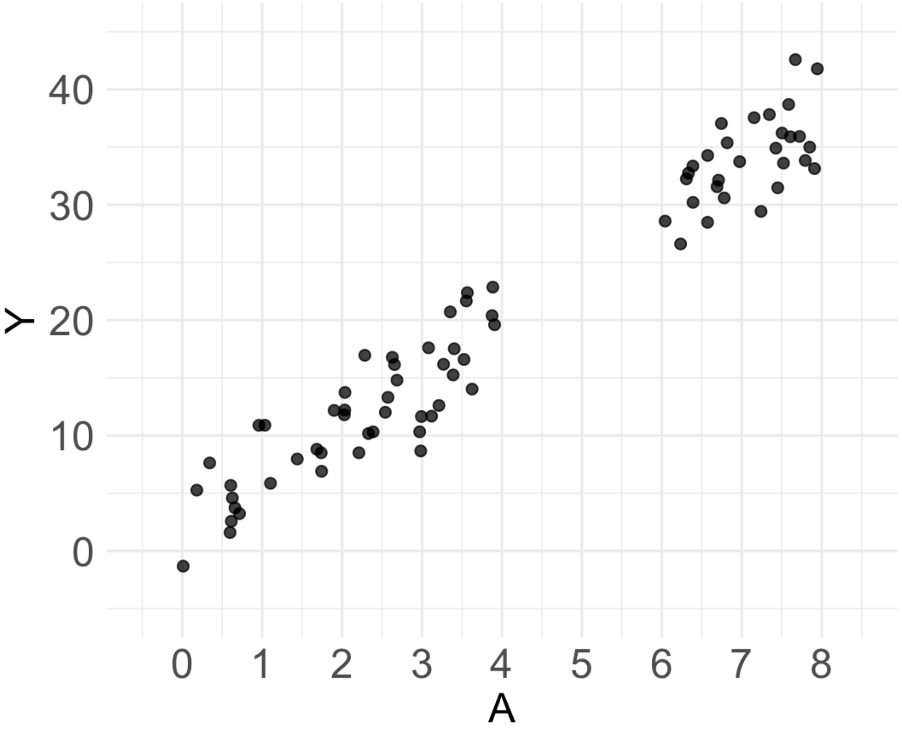

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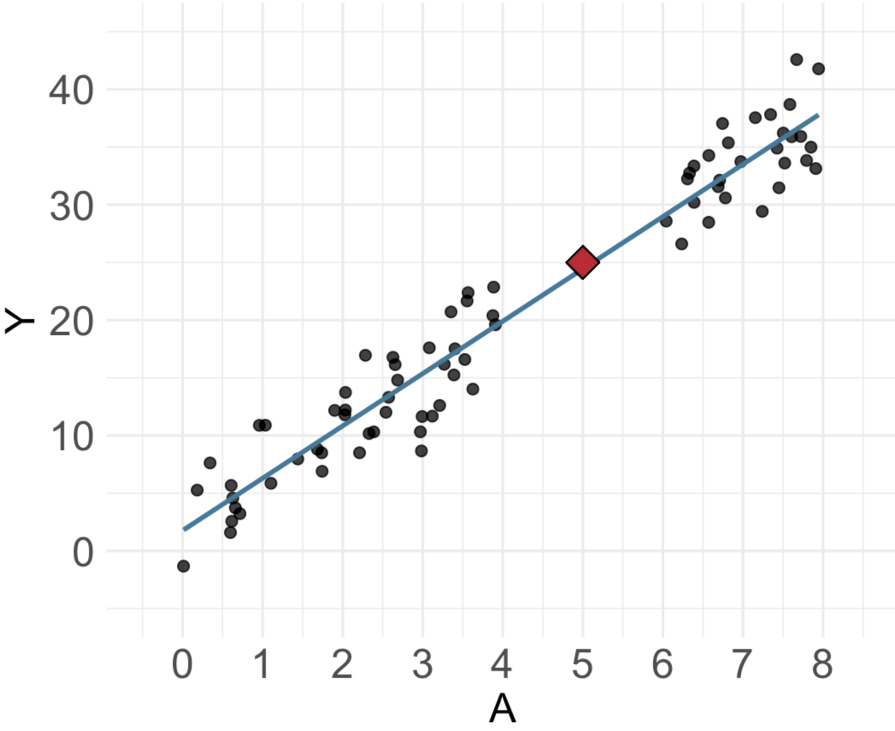


Estimation

Identification Assumptions → **Modeling Assumptions** →

Causal Estimand

$$\underbrace{E[Y^{a=1}|L] - E[Y^{a=0}|L]}_{\text{Counterfactual (unobservable)}}$$

Statistical Estimand

$$\underbrace{E[Y|A=1,L] - E[Y|A=0,L]}_{\text{Factual (observable)}}$$

Estimate

$$\underbrace{E[Y|\widehat{A=1},L] - E[Y|\widehat{A=0},L]}_{\text{Number}}$$

- Regression analysis
  - → Making educated guess about conditional expected values
    - By borrowing information from other observations (model interpolation)
      - With modeling assumptions (e.g., linearity, no product terms)
        - CATE or ATE?
    - Pure prediction task after identification
      - Causal inference meets prediction!

> ⚠ **You get meaningless “numbers” with:**
>
> - Good estimation but bad identification
> - Good identification but bad estimation

Modeling in Practice

- Simple model with stronger assumptions (most parametric regression)
  - → More efficient but more biased
- Complex model with weaker assumptions
  - → Less biased but less efficient

Solutions to the bias-variance trade-off

- Doubly-robust estimation (e.g., TMLE) [1]
- Non-parametric & data-adaptive approach (e.g., machine learning, SuperLearner) [2]
  - → Generally better performance [3]
    - But potentially small gain with small and less complex data
      - Social epi?

Causal Inference: Big Picture

Data

Identification Assumptions

Modeling Assumptions

$$\underbrace{E[Y^{a=1}] - E[Y^{a=0}]}_{\text{Counterfactual (unobservable)}} \quad \underbrace{E[Y|A=1] - E[Y|A=0]}_{\text{Factual (observable)}} \quad \underbrace{E[\widehat{Y|A=}1] - E[\widehat{Y|A=}0]}_{\text{Number}}$$

Causal Estimand ⟩⟩⟩ **Statistical Estimand** ⟩⟩⟩ **Estimate**

Interpretation

- Unjustified fancy methods, unreasonable critiques
  - → "Propensity scores can strengthen confounding adjustment!"
    - Identical conditional exchangeability assumption (Shiba and Kawahara 2021)
  - → "Never use data-driven methods in causal research. Use theories!"
    - Could be useful for estimation.

! Always distinguish these steps when you learn, read, and respond

Summary 2

Data

Identification Assumptions

Modeling Assumptions

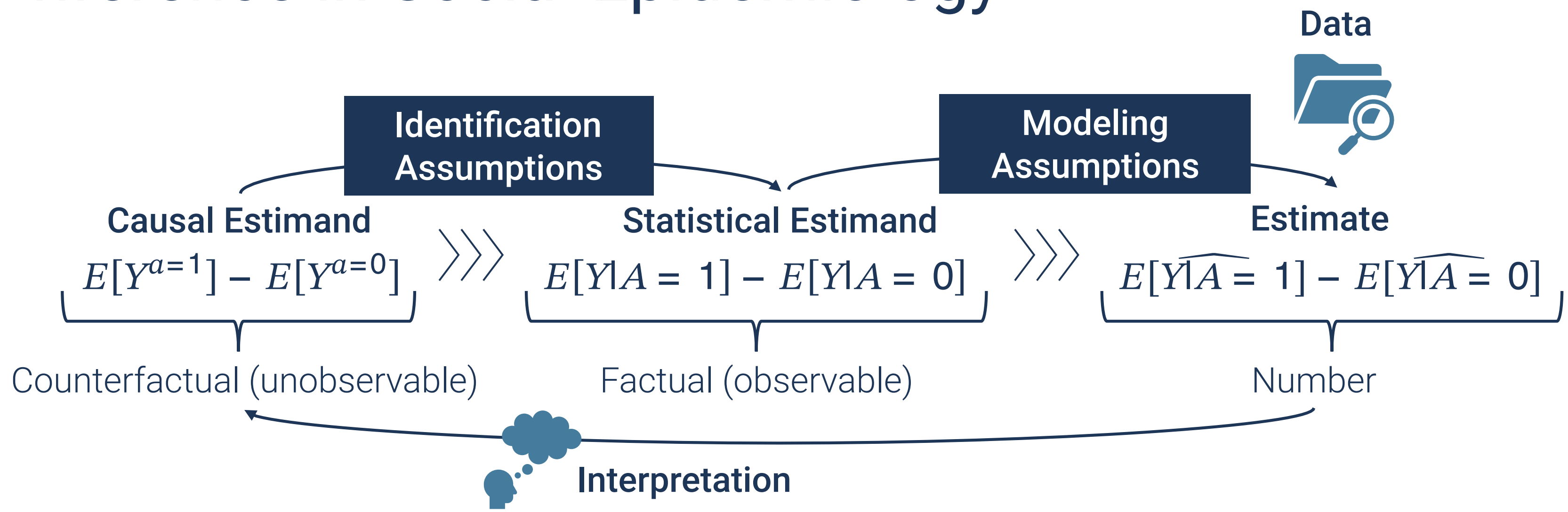
Causal Estimand ⟫ **Statistical Estimand** ⟫ **Estimate**

$$\underbrace{E[Y^{a=1}] - E[Y^{a=0}]}_{\text{Counterfactual (unobservable)}} \quad \underbrace{E[Y|A=1] - E[Y|A=0]}_{\text{Factual (observable)}} \quad \underbrace{E[\widehat{Y|A=}1] - E[\widehat{Y|A=}0]}_{\text{Number}}$$

Interpretation

- Three steps of causal inference
    1. Specifying causal estimand
    2. Identification with assumptions
        - → exchangeability, positivity, consistency
    3. Estimation via statistical models
- Interpretation & data

Causal Inference in Social Epidemiology



- Data (Week 4)
  - → Measuring intangible & complex social constructs
- Causal Estimand (Week 5)
  - → Multi-level, distributional, population impacts
- Identification
  - → Exchangeability (Week 3)
  - → **Consistency** (Today)

Consistency Assumption for Social Exposures

- $Y^a = Y$ when $A = a$
  - → Intuition: "The observed outcome Y of the exposed is equal to what the outcome would have been had they been exposed"
  - → Corresponding interventions?

⚠ **Challenges**

- **Multiple versions (compound treatment)**
  - → Crude definition (social participation, neighborhood vulnerability)
    - Often harder to achieve the treatment-variation irrelevance assumption
- **Nonmanipulable?**
  - → What interventions would correspond to the "effect of race"?

Galea-Hernan 2019 AJE Article

JOURNAL ARTICLE EDITOR'S CHOICE

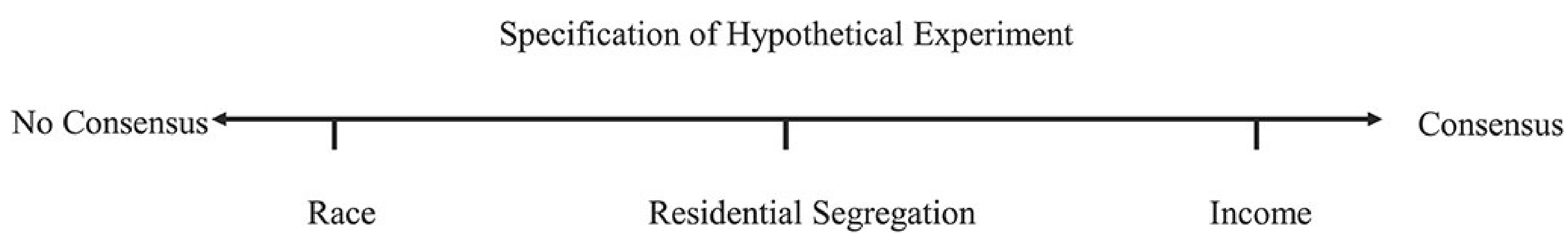
Win-Win: Reconciling Social Epidemiology and Causal Inference FREE

Sandro Galea, Miguel A Hernán

American Journal of Epidemiology, Volume 189, Issue 3, March 2020, Pages 167–170,
https://doi.org/10.1093/aje/kwz158

Published: 03 October 2019 **Article history** ▾

Claim: Move to the right of the proposed spectrum of social exposures



Responses to Galea & Hernan (2019)

Responses concerning the consistency assumption for social exposures

1. **Robinson and Bailey** (2020)
    - Causal effect estimation is not the only goal of social epi
        - → (Preliminary) evidence to "establish or rule out causation"
            - "Can genetic differences explain racial disparities?"
        - → *Changes are driven by basic beliefs & narratives about causality, cultural values, attribution of moral responsibility rather than precise quantitative estimates*
    - Can fall into the "reductionism trap"
        - → Finding "spiders" vs victim blaming (Krieger 1994)

2. **Jackson and Arah** (2020)
    - System preserving
    - Systems and effect

Responses to Galea & Hernan (2019)

3. **VanderWeele** (2020)
    - Quantitative potential-outcome framework for social exposures on the “left end”

Alternative approaches for ill-defined social exposures

1. Redefine the exposure (to make modifiability more explicit)
2. Changing interpretation
3. Shifting the target of interventions

Approach 1: Redefining Exposures

- The effect of “race”
  - → Hypothetical intervention?
  - → Changing skin color? Label assigned at birth?
  - → **Race as a social construct**
    - Racism, discrimination, perceptions
      - e.g., White-sounding names got more interviews (Bertrand and Mullainathan 2004)
- Not saying we should avoid studies of “race”
  - → Quantitative causal inference as a tool is more useful for well-defined exposures

Approach 2: Changing Interpretation

- Social participation $A = 1$ (vs no participation: $A = 0$)
  - → Version 1: ≥ once a week ($K(1) = 1$)
  - → Version 2: less often ($K(1) = 2$)

Possible interpretations

1. **Effect of "assigning" a compound treatment**
   - $E[Y^1 - Y^0]$
2. **Weighted average of version-specific effects Y^{a,k^a}**
   - $\sum E[Y^{a,k^a}|A = a, K^a(a) = k^a]Pr[K^a(a) = k^a|A = a]$
     - → *Proposition 3* in (VanderWeele and Hernan 2013)
       - Additional conditioning on L in observational studies
3. **Effect of assigning a random draw from versions K**
   - From observed distribution among $A = 1$ vs $A = 0$
     - → & further conditioning on $L = l$ if confounding was adjusted

Approach 3: Shifting The Target of Interventions

$$\text{Race} \overset{?}{\dashrightarrow} \text{SES} \longrightarrow Y$$

- The effect of SES on mitigating the racial disparity (VanderWeele and Robinson 2014)
  - → What is the remaining association between race and the outcome, after fixing SES?
  - → Often viewed as an extension of **mediation**
    - ▫ But does not require the conceptualization of an effect for the "exposure"

Summary 3

- **Consistency assumption is often challenging for social exposures**
  - → Unclear actions, lack of transportability, ill-informed study design
- **In general, aim for well-defined & manipulable exposures**
  - → Imagine hypothetical interventions
  - → (Perhaps less intuitive) interpretation available even when there are multiple versions
- **But ask fundamental questions: What are the spiders that are producing inequalities?**
  - → Cutting the “web of causation” isn’t sufficient
  - → The closer we get to the spiders, the less well-defined?
    - What is the goal of your “causal analysis”? Effect estimation or explanation?

Assignment

- Problem set #2 (due September 22 at 9:55am)
- Reading quiz

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